HW1 UserName Validation Test Cases

Target rule: 4-32 characters; the first character is alphabetic; later characters are alphanumeric; -, _, ., and & are permitted only between alphanumeric characters. Error positions below are one-based, matching the GUI alert.

ID	Input	Purpose	Expected	Expected message	Error position	Actual	Result
TC01	<empty>	Reject missing input	Invalid	The input is empty.	1	Invalid	**PASS**
TC02	1abc	Reject a digit first	Invalid	Must start with A-Z or a-z.	1	Invalid	**PASS**
TC03	Ab12	Accept minimum length	Valid	No error	-	Valid	**PASS**
TC04	a-b_c.d1	Preserve original separators	Valid	No error	-	Valid	**PASS**
TC05	one&two	Accept new & separator	Valid	No error	-	Valid	**PASS**
TC06	Ab1	Reject below minimum	Invalid	At least 4 characters.	4	Invalid	**PASS**
TC07	A + 31 digits	Accept 32-character boundary	Valid	No error	-	Valid	**PASS**
TC08	A + 32 digits	Reject 33 characters	Invalid	No more than 32 characters.	33	Invalid	**PASS**
TC09	one&	Reject trailing separator	Invalid	Separator must be followed by UNChar.	5	Invalid	**PASS**
TC10	one&&two	Reject consecutive separators	Invalid	Separator must be followed by UNChar.	5	Invalid	**PASS**
TC11	one@two	Reject unsupported character	Invalid	Only permitted characters and separators.	4	Invalid	**PASS**
TC12	one two	Reject embedded space	Invalid	Only permitted characters and separators.	4	Invalid	**PASS**

Execution summary: 12 of 12 cases passed. The full automated suite also exercised each separator independently, mixed separators, a normal alphabetic UserName, and adjacent different separators; 18 of 18 automated checks passed.